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Data Residency and Hosting Model Document: VI Business (India)

1. Executive Purpose and Document Scope

The Data Residency and Hosting Model Document codifies the physical locations, logical boundaries, and architectural patterns governing customer data processed by VI Business (India).

With the enactment of the Digital Personal Data Protection (DPDP) Act 2023, and strict sectoral guidelines from the Reserve Bank of India (RBI) and Insurance Regulatory and Development Authority of India (IRDAI), this document serves as the canonical proof of sovereign data compliance. It equips Sales, Pre-sales, and Enterprise Legal teams to accurately address strict regulatory questionnaires during RFP evaluations.

2. Definitions, Assumptions, and Applicability

Definitions:

- **Customer Payload:** Raw data transiting the network or passing through security firewalls (e.g., body of an email, proprietary ERP data).
- **Metadata / Telemetry:** Data about the payload (e.g., source IP, destination IP, bandwidth consumed).
- **DPDP:** Digital Personal Data Protection Act 2023.

Assumptions:

- **Total Sovereign Isolation:** No enterprise customer metadata, logs, or billing information processed by VI Business India core platforms leaves the geographic boundaries of the Republic of India.
- **Lawful Interception:** All hosting models described below comply with Department of Telecommunications (DoT) mandates for lawful interception and monitoring by Indian state agencies.

Applicability:

- **Target Audience:** Bid Writers, Solution Architects, Legal Counsel, and Enterprise Auditors.
- **Scope:** Managed Network Services, Security Operations Center (SOC), Enterprise Mobility (IoT/cAPN), and API Gateways.

3. The India Data Residency Position

VI Business explicitly guarantees 100% data residency within India for all managed service control planes and telemetry storage.

Data Category	Storage Location	Archival Storage Location	Offshore Access Permitted?

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Billing & Financial Ledgers	Mumbai (NESCO) Tier-III DC	Pune Tier-III DC (Replication)	Strict No
NOC/SOC Telemetry Logs	Pune Tier-III DC	AWS ap-south-1 (Mumbai)	Strict No
IoT / cAPN Routing Data	Dual Core: Mumbai & Pune	Dual Core: Mumbai & Pune	Strict No
Customer API Credentials	Mumbai (NESCO) Tier-III DC	Pune Tier-III DC (Replication)	Strict No

4. Hosting Options and Architectural Patterns

VI Business offers flexible hosting models dictating how the administrative control planes (like SD-WAN Orchestrators) are deployed. Note: VI Business does *not* inspect Customer Payload unless explicitly contracted as a Deep Packet Inspection (DPI) managed security service.

4.1 Multi-Tenant SaaS (Default Pattern)

- **Architecture:** The customer tenant is logically segregated within a shared orchestration cluster hosted in the VI Business Mumbai Tier-III core.
- **Use Case:** Standard SD-WAN deployments, typical Enterprise Mobility management, and shared SOC ingestion.
- **Positioning:** Highly scalable, cost-effective, and fully redundant. Logical isolation is verified by annual SOC 2 Type II audits.

4.2 Dedicated Instance (Private Cloud)

- **Architecture:** A chemically isolated Virtual Machine (VM) cluster hosted in the VI core, entirely dedicated to a single customer entity. NO shared databases.
- **Use Case:** Highly regulated BFSI entities demanding absolute physical compute isolation.
- **Positioning:** Requires Deal Desk approval as it alters standard licensing rules and introduces premium CapEx surcharges (>INR 25 Lakhs OTC).

4.3 Hybrid / Cloud Edge

- **Architecture:** The control plane resides in the VI Mumbai Data Center, but the virtualized CPE (vCPE) processing the customer payload is spun up natively inside the customer's AWS (ap-south-1) or Azure (Central India) VPCs.
- **Use Case:** Cloud-first enterprises migrating away from on-premise hardware data centers.

5. Customer Data Boundary Controls

To ensure multi-tenant security, the following logical boundary controls are unconditionally enforced:

1. **Role-Based Access Control (RBAC):** VI Business NOC/SOC engineers operate under the Principle of Least Privilege. Access to customer tenants requires explicit Just-In-Time (JIT) provisioning tied to active ServiceNow incident tickets.
2. **Encryption at Rest:** All stored telemetry, Syslog entries, and firewall rule-bases are encrypted using AES-256 standard. Keys are rotated every 90 days via an isolated Hardware Security Module (HSM).

3. **Data Masking:** PII (such as employee mobile numbers within the EMM portal) is masked for VI L1/L2 engineering staff. Unmasking requires L3 authorization and a logged justification code.

6. Backup and DR Location Rules

Data survivability is inextricably linked to data residency.

- **Primary Active Data:** Stored in the Mumbai (NESCO) core.
- **Disaster Recovery (DR) Sync:** Asynchronously replicated via dark fiber to the Pune Tier-III facility.
- **Immutable Offline Archival:** Long-term compliance logs (retained for 12 months per DoT mandate) are stored in AWS S3 buckets strictly locked to the `ap-south-1` (Mumbai) region.

7. Constraints, Exclusions, and Dependencies

- **Exclusion regarding Payload:** If a customer utilizes a VI-provided internet lease line (ILL) to send unencrypted data to an offshore server in the US or Europe, VI Business has no control over that transit routing. Data residency guarantees apply *only* to the VI-managed control platforms, SIEM telemetry, and localized cAPN IoT routing.
- **Auditor Verification:** Customers requiring onsite physical inspections of the Mumbai or Pune data centers must request access 30 days in advance, subject to non-disclosure agreement (NDA) clearance.

8. Proposal-Use Guidance

- **RFP Use:** When an RFP asks, "Will our data ever leave India?", copy Section 3 verbatim into the compliance matrix.
- **Multi-tenant Defense:** Customers often blindly demand "Dedicated Hosting" without realizing the cost. Use Section 4.1 to defend the security of the Multi-Tenant model by citing "Logical Segregation validated by SOC 2 Type II."

9. Governance and Ownership

- **Owner:** Enterprise CISO (Rajesh Kumar)
- **Reviewers:** Head of Enterprise Legal, National Head of Operations
- **Update Cadence:** Evaluated annually, or immediately upon the release of new DoT guidelines or amendments to the DPDP Act 2023.